

DBMS – PRODUCT AND CATEGORY MANAGEMENT

ZEPTO E-COMMERCE

1. Database Design:

Category Table:

Attribute	Description
CATEGORY_ID	Unique category ID
CATEGORY_NAME	Category name
DESCRIPTION	Category description

Product Table:

Attribute	Description
PRODUCT_ID	Unique product ID
PRODUCT_NAME	Product name
PRICE	Product price
STOCK	Available stock
CATEGORY_ID	Category ID
DESCRIPTION	Product description

2. Define Primary and Foreign Key Relationships:

Category → Product

- CATEGORY.CATEGORY_ID → Primary Key
- PRODUCT.PRODUCT_ID → Primary Key
- PRODUCT.CATEGORY_ID → Foreign Key referencing CATEGORY.CATEGORY_ID
- Relationship: One-to-Many (1:M)

One category can contain many products.

3. Table Creation:

Create Category Table:

```
CREATE TABLE CATEGORY(  
2     CATEGORY_ID NUMBER PRIMARY KEY,  
3     CATEGORY_NAME VARCHAR2(30),  
4     DESCRIPTION VARCHAR2(50)  
5 );
```

Table created.

Create Product Table:

```
SQL> CREATE TABLE PRODUCT(  
2     PRODUCT_ID NUMBER PRIMARY KEY,  
3     PRODUCT_NAME VARCHAR2(30),  
4     PRICE NUMBER(10,2),  
5     STOCK NUMBER,  
6     CATEGORY_ID NUMBER,  
7     DESCRIPTION VARCHAR2(50),  
8     FOREIGN KEY(CATEGORY_ID) REFERENCES CATEGORY(CATEGORY_ID)  
9 );
```

Table created.

4. Data Insertion:

Category Records:

```
SQL> INSERT INTO CATEGORY VALUES(1,'FRUITS','Fresh fruits');
```

1 row created.

```
SQL> INSERT INTO CATEGORY VALUES(2,'VEGETABLES','Fresh vegetables');
```

1 row created.

```
SQL> INSERT INTO CATEGORY VALUES(3,'DAIRY','Milk and dairy products');
```

1 row created.

```
SQL> INSERT INTO CATEGORY VALUES(4,'SNACKS','Snacks and biscuits');
```

1 row created.

```
SQL> INSERT INTO CATEGORY VALUES(5,'BEVERAGES','Juices and soft drinks');
```

1 row created.

Product Records:

```
SQL> INSERT INTO PRODUCT VALUES(101,'APPLE',120,50,1,'Fresh red apples');

1 row created.

SQL> INSERT INTO PRODUCT VALUES(102,'BANANA',60,80,1,'Fresh bananas');

1 row created.

SQL> INSERT INTO PRODUCT VALUES(103,'TOMATO',40,70,2,'Fresh tomatoes');

1 row created.

SQL> INSERT INTO PRODUCT VALUES(104,'MILK',35,100,3,'Fresh milk');

1 row created.

SQL> INSERT INTO PRODUCT VALUES(105,'BISCUIT',30,60,4,'Cream biscuits');

1 row created.
```

5. Data Retrieval:

Display Category Records:

CATEGORY_ID	CATEGORY_NAME	DESCRIPTION
1	FRUITS	Fresh fruits
2	VEGETABLES	Fresh vegetables
3	DAIRY	Milk and dairy products
4	SNACKS	Snacks and biscuits
5	BEVERAGES	Juices and soft drinks

5 rows selected.

Display product records

```
SELECT * FROM PRODUCT;
```

PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK	CATEGORY_ID
101	APPLE Fresh red apples	120	50	1
102	BANANA Fresh bananas	60	80	1
103	TOMATO Fresh tomatoes	40	70	2
104	MILK Fresh milk	35	100	3
105	BISCUIT Cream biscuits	30	60	4

5 rows selected.

6. Data Updation:

Update Product Price:

```
SQL> UPDATE PRODUCT
2 SET PRICE = 130
3 WHERE PRODUCT_ID = 101;

1 row updated.
```

Update Product Stock:

```
SQL> UPDATE PRODUCT
2 SET STOCK = 90
3 WHERE PRODUCT_ID = 104;

1 row updated.
```

7. Data Deletion:

Delete Product Record:

```
SQL> DELETE FROM PRODUCT
| 2 WHERE PRODUCT_ID = 105;

1 row deleted.
```

8. Category-Wise Product Report:

```
SQL> SELECT C.CATEGORY_NAME, P.PRODUCT_NAME, P.PRICE, P.STOCK
| 2 FROM CATEGORY C
| 3 JOIN PRODUCT P
| 4 ON C.CATEGORY_ID = P.CATEGORY_ID;
```

CATEGORY_NAME	PRODUCT_NAME	PRICE
FRUITS	APPLE	130
FRUITS	BANANA	60
VEGETABLES	TOMATO	40

CATEGORY_NAME	PRODUCT_NAME	PRICE
FRUITS	APPLE	130
FRUITS	BANANA	60
VEGETABLES	TOMATO	40

CATEGORY_NAME	PRODUCT_NAME	PRICE
FRUITS	APPLE	130
FRUITS	BANANA	60
VEGETABLES	TOMATO	40

CATEGORY_NAME	PRODUCT_NAME	PRICE
FRUITS	APPLE	130
FRUITS	BANANA	60
VEGETABLES	TOMATO	40

9. Display Products Under Fruits Category

```
SQL> SELECT P.PRODUCT_NAME, P.PRICE, P.STOCK
2  FROM PRODUCT P
3  JOIN CATEGORY C
4  ON P.CATEGORY_ID = C.CATEGORY_ID
5  WHERE C.CATEGORY_NAME = 'FRUITS';
```

PRODUCT_NAME	PRICE	STOCK
APPLE	130	50
BANANA	60	80

10. Category-Wise Product Count

```
SQL> SELECT C.CATEGORY_NAME, COUNT(P.PRODUCT_ID) AS TOTAL_PRODUCTS
2  FROM CATEGORY C
3  LEFT JOIN PRODUCT P
4  ON C.CATEGORY_ID = P.CATEGORY_ID
5  GROUP BY C.CATEGORY_NAME;
```

CATEGORY_NAME	TOTAL_PRODUCTS
FRUITS	2
VEGETABLES	1
DAIRY	1
SNACKS	0
BEVERAGES	0